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A Review of Star schema and Snowflakes Schema

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Abstract. In the new age, digital data is the most important source of acquiring knowledge. For this purpose, collect data from various sources like websites, blogs, webpages, and most important databases. Database and relational databases both provide help to decision making in the future work. Nowadays these approaches become time and resource consuming there for new concept use name data warehouse. Which can analyze many databases at a time on a common plate from with very efficient way. In this paper, we will discuss the database and migration from the database to the data warehouse. Data Warehouse (DW) is the special type of a database that stores a large amount of data. DW schemas organize data in two ways in which star schema and snowflakes schema. Fact and dimension tables organize in them. Distinguished by normalization of tables. Nature of data leads the designer to follow the DW schemas on the base of data, time and resources factor. Both design-modeling techniques compare with the experiment on the same data and results of applying the same query on them. After the performance evaluation, using bitmap indexing to improve the schemas performance. We also present the design modeling techniques with respect to data mining and improve query optimization technique to save time and resource in the analysis of data.

Keywords: Data Warehouse, Data Mart, Star Schema, Snow Flakes, Relational database management system RDBMS.

1 Introduction

Data Warehouse is one of the emerging and mainly focused technology and nowadays being used almost in every enterprise to increase business performance. DW is a system that fetches data from different sources within an organization, to deduce some reports or analytical information and predict some patterns for further research. To organize data in DW, different schemas can be handy like star schema (SS) and snowflake (SF) schema. Both of the schemas use around the globe in the process [1] of Extract, Convert and Load (ETL). ETL is mainly a three-step process in the making of a data warehouse. There are many things which are to be compared between to both of these, star

and snowflake schema before their usage. However, their roots are same or they originate from the same base but they have many differences when it comes to their usage, performance, storage limitations, etc. Both these schemas use fact table as the main table which is linked with the dimensions or the dimension tables, that make the arrangement of tables to be called as data marts and data warehouses, on which queries are applied and these schemas then further are also responsible for query performance which can grow really complex [2]. The importance of both these schemas must be reviewed to make sure the performance of DW and to ensure productive and effective results for the enterprise's growth. As there is also not much work available on the specific domain of these schemas importance and considering its use nowadays that is growing exponentially, so this will definitely be a contribution to the knowledge. We had a database to store data files and use this according to our need. However, the access of data from the database has some limitations that restrict the user to access the open way. We use some programming techniques to access this data according to our requirement. In the new age after the Relational model's data can be used for a lot of ways to get the results [3]. For this purpose, we use data warehouses. DWs use to manage big data for our business purposes.

1.1 Database and Relational Database

Collection of data that holds the tables called database [4]. Database play a vital role in the world to store data with efficient way but in the simple database have some discrepancies that cannot meet in the new age needs therefor introduce RDBMS. In this way data store more efficient way and can access more easy ways. When we use the large database, we want to real-time analysis for this purpose speed, and resources consumption go very high therefor to hold these aspects we use DWs [5].

In Fig:1 represent the process of transforming the data from a database towards the knowledge discovery process. For this purpose first, collect data from the OLAP system then refine it for the specific purpose and put in the DW. This data cleansing and remove the ambiguity for fit in the DW. Select the specific data and design data marts of them. Different data marts design for different purpose of analysis for example sales analysis, products impact and price effect on the business area. For all analysis design individual data marts [6].

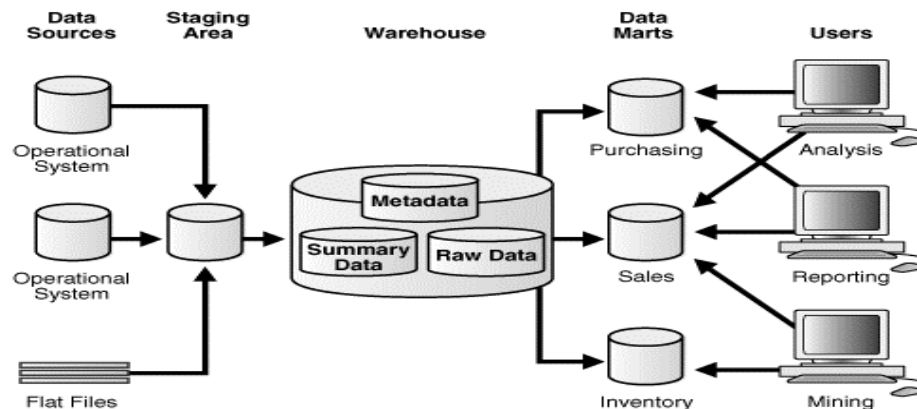


Fig. 1. Data Transform to Data Marts via DWs [2]

1.2 Data Warehouse

In the database, data store simultaneously time by time and become big to bigger. For the sake of analysis and management, become difficult to handle this data. to overcome this problem we felt the need for data warehouses and DataMart. A data warehouse is the collection of many databases that are combined for the analysis purpose via quantitatively or qualitatively. This enables us to get the required test data for further process [3]. DWs like a database that uses the information for reports in a different way to establish new information from previous. DWs is not a full database also can be the specific relevant data that is beneficial for our industry or company [7]. DWs holds the raw data that will not be part of our research.

1.3 Data Marts

In Data warehouse whole databases, data stored that is memory consuming. We create DWs to combine data from many sources that need to store memory in terabytes. In "Corporate America " data Marts use in the implementation of DWs [8]. Therefore a more abstract form of DWs, we use Data Marts. DMts contain the specific domain data that captured for analysis on a specific topic [9]. For example, we create a DWs that contain the data of laboratory in which store previous 20 years data but we extract only 5 years data for a specific disease prediction or growth rate of disease in last 5 years [5].

1.4 Data Mining

Data mining is the next step of DMts in which we make patterns from data. use organized data to analyze using various methods. it is a computational process in which use organized data to get future predictions and some exploring knowledge with a different way. It basically provides a certain kind of intelligence for the company. Set of queries

are run the result of which is further considered to obtain help for the decision making [10].

2 Problem statement

A data warehouse can be managed in many ways but two commonly used types are star schema (SS) and snowflakes schema (SF). The questions are that how can distinguish which schema is best in which DW because in the industry time and resources have much value in the eye of financers. For this purpose, we analyze these schemas with structured efficiency. First, we describe the working of a star schema in which also include an example that helps us for results after this describe SF schema. Comparison of these schemas helps us to lead the best approach for future work.

3 Literature Review

In current literature, we analyze some DW techniques that use to knowledge discovery from selected databases. Collection of data mine in these techniques and fetch out the useful information as the result.

3.1 Design Modeling Techniques

For the selection of exact schema for the analysis, various modeling techniques are used on both the relational database and DWs. The creator ought to have a clear comprehension of the relevance and inability of the techniques. Two main models follow to design schema also in relational database and DWs are Data Model and Process Model [11].

Process Model:

Process model when dealing with data warehouse, it can not be suitable because of the reason that it deals with requirement assumptions which obviously is not suitable for the DW. As it treat with more requirement assumptions so it is better to be used for some operational environment and because of this the other model which is data model comes in. Following are the domains where process models are applicable [12-14].

1. Functional breakdown.
2. Environment-level diagram.
3. Data flow diagram.
4. Operational chart.
5. State-run diagram.
6. Hipo chart.
7. Pseudocode [1].

Data Model:

Now modeling of data is the one considered compared to the process model for the data warehouse. It basically carries two main techniques which are Entity relational and dimension modeling.

Dimensional modeling.

Dimension modeling focus on the behavior of fact table in any schema because in the fact table whole PKs lies as the foreign key that shows exactly one record for specifying index this behavior also called the star schema. In the expression of element relationship compelled adaptation are utilized for creating dimensional DW. In star schema accentuation is on one table (Fact table) with the basic principle depicting the association, not at all like conventional schema no specific table has a complete thought.

3.2 Evaluation of schemas

As per after some time the manner in which things being done changes, the thought behind data model structure of related application may encounter intense change. Measurement changes, instant changes, actuality changes, level changes, property changes, limitation changes and quality changes are conceivable emerging changes in the outline plan particularly for DW. The outline advancement challenges are everlasting and relevant to each database for all intents and purposes conceivable [15]. Data warehouse center is created iteratively with the goal that key choice, later on, could be conceivable.

3.3 Star Schema

In the ETL process that is follow to design a DW, dimension modeling is used that is done on different models, one of which is star schema. As the name suggests “star”, its shape is actually like SS consists of dimensions, which are in dimension tables and a central fact table that carries facts [16, 17]. Fact table carries in it the PKs that lies in dimension tables add with the name of foreign keys, which is the means of connectivity between dimension and fact table. Dimension tables have qualitative data and as discussed above with the FK fact table contains measures that can be summed to analyze or to carry some process [1].

Properties of a star schema

- a. Dimensions tables pass the PK add with the name of the FK in fact or base table.
- b. Firstly, it contains tables that distinguished with, mainly fact table and dimension tables.
- c. Star schema usually not in normalized form except the fact table in the schema.
- d. Whenever tables are not normalized it obviously due to redundancy memory will need more, more space will be required.
- e. Star schema use all dimensions table’s PKs, therefore, fact table look less complex instead of snowflakes.
- f. DE normalized means lesser tables and lesser tables mean less complex queries. Queries that are used in star schema to access data are “star join queries”.

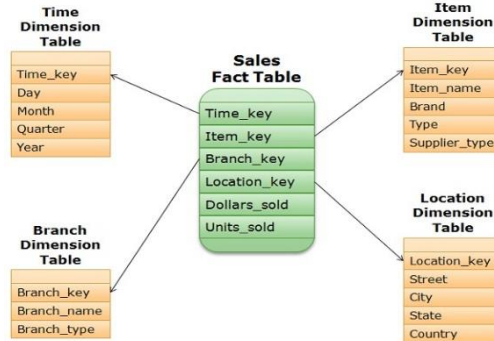


Fig. 2. Star Schema Represent in Pictorial [18].

To explain the results for the query performance of star schema, considering the following Example.

Table 1. Required Specifications for the Experiment [19].

Physical Memory	Storage Disk	Operation System	RDBMS
8 Gb	1 TB	Win 2008 Server	MS SQL SERVER 2008 R2

Environment under which tests were performed, we required 8GB RAM and 1TB hard disk for storage and use Operating system Windows 2008 Server or 2012, SQL server 2008. These hardware specifications are not hardcode for these schemas these are only for the purpose of the experiment.

Table 2. Star Schema Query [1].

```

SELECT  d1.att2,    d3.att2,    d1.att1,
SUM(f.measure1)
from Fact1 f inner join dim1 d1 on
d1.dimkey1 = f.dimkey1
GROUP BY d1.att1

```

Star schema used in the example has one fact (main) table and three dimensions (connecting) tables. In Table.2 query used for the SS.

Things which we are going to judge through this query used will be the time taken for execution and memory used by it in the query d1 dimension table 1 and column att1 and att2 with dimension Table 3 were exist att2 values and accrue these entire columns and sum their values. Results will be obtained just by applying the query to the designed schema. The results obtained are:

Table 3. Star Schema Query Performance [1].

Query	Memory Occupied by process	Real time Query execution in seconds	DW size
Star Join	72%	21s	0.78 GB

Compensations of Star Schema:

- In a SS, not all tables are normalized but only till first normal form.
- In Star schema, the execution time of the query is fast when we use large data.
- Because a star schema database has a small number of tables and clear join paths, Small single-table queries, usually of dimension tables, are almost instantaneous.
- SS design is easy to follow and especially, with respect to the joined query, which tends with only through the fact table. These joints help to boost the performance of schema.
- Only simple queries use because all tables less complex therefore time to run and output result show less time.

3.4 Snowflakes Schema

Like SS, the name of snowflake schema also comes through its resemblance with a snowflake. Like star schema, it is also used in the ETL process and at the same level. There is a choice between using either of them. Unlike star schema, snowflake schema consists of three types of tables, which are, fact table, dimension tables, and sub-dimension tables. Working of the fact table is the same as it carries Foreign keys as of dimension tables as FK and contains its own other values. The fact or base table connected with dimension tables and dimension tables more time connect to normalize dimension tables through keys. Due to sub-dimensions or the splitting of dimensions into the sub-dimensions concept of hierarchy is introduced in snowflake schema [19].

Snowflake apply on a simple database environment also works with different outsider devices and administrations, for example, Informatics, Looker or Tableau notwithstanding, it likewise furnishes the likelihood to collaborate with the framework utilizing only an internet browser [20].

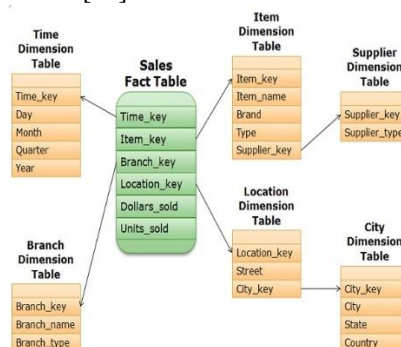


Fig. 3. Snowflakes Schema [21].

Properties of Snowflake Schema:

- a. It contains three types of tables, fact table, dimension tables, and sub-dimension tables.
- b. Sub dimension tables constructed by splitting or normalizing dimension tables, so it can be call snowflake schema is in normalized form.
- c. As this schema is, normalize so data is not redundant, repetitions are not present and values are atomic.
- d. As the redundancy removed from the tables so lesser memory will be required, means a decrease in storage space required.
- e. SF structure more complex because many tables arranged when we normalize database till 3rd normal forms.
- f. One problem is there which can be considered, due to increase in a number of tables, querying the data from these tables will be a tough job because linking tables from the sub-dimension of one table to sub-dimension of another table can be complex [22].

Advantages of snowflakes:

A big online industry uses snowflakes schema due to some important benefits.

Relational : This schema support all transactions of major plate forms like SQL, ANSI, and ACIDS. The big benefit is that u can migrate from one to another platform without a big change.

Semi-Structured: The query performance increase with the built-in functions that help for navigating, destruction and organize semi-structure data that use in nested form. Also with the help of JSON and Avro. Fewer operations perform due to the auto discovery of schema and storage operations less perform with respect to efficiency and save the efforts.

Elastic: Usually called the portable schema because of scale and resource independent schema. Port to another schema without damaging the availability and queries concurrent.

Highly Available: Snowflakes schema take less time to recover and rare chances to failure there for tolerance node and cluster less effect the performance. Without down can change the hardware.

Durable: Snowflake can be called durable due to some properties of cross-region backup. Another backup performed this schema with the help of cloning undrop.in this way any damage cannot highly affect the schema.

Cost-efficient: Snowflake highly cost-efficient because fewer resources required and compressed data stored.

Secure: Whole data that store in the schema is end to end encrypted include all traffic over the network and temp files. Additionally, access control also help the user to protect the data.

Performance: Discussing about star schema and snowflake schema , with bigger fact tables which means containing greater amount of data or the FK in fact tables can reduce the performance of star schema or can be said the lesser the FK in fact table and more the partitioned is the table the faster and more efficient it will work [19]. When using star and snowflake schema in a similar environment it is obvious (considering the example from):

- It is evident that when dimension tables are bigger it is good to use snowflake schema because it is in normalized form the dimension tables are partition and hence it reduces the size leading to less memory consumption.
- Query processing time increases when using star schema with large data.

Table 4. Required Specifications for the Experiment [1].

Physical Memory	Storage Disk	Operation System	RDBMS
8 Gb	1 TB	Win 2008 Server	MS SQL SERVER 2008 R2

The example of SS taken consists of one fact (Main) table with three dimensions (connecting) tables and further having two sub-dimensions. The query used to test the performance of snowflake schema is the following:

Table 5. Snowflakes Schema Query [1].

```
SELECT fact2.fact2_att1,
fact2.fact2_att2,fact2.fact2_att3,
d11.d11_att1, fact2.fact2_att3,
fact2.fact2_att3
FROM d11 LEFT JOIN d1
ON d11.d11_att1 = d1.d11_att1 LEFT JOIN
fact2 ON d1.d11_att1 = fact2.fact2_att1
GROUP BY fact2.fact2_att1;
```

Same parameters will be checked for snowflake schema as they were checked for star schema and the results obtained are also from directly querying through schema no optimizations were done, following are the results.

Table 6. Snowflakes Schema Performance [1].

Query	Memory Occupied by process	Real time Query execution in seconds	DW size
Snowflake Join	49%	17s	0.61 GB

For obvious reasons, the size of the data warehouse will a bit smaller due to the fact that snowflake schema is in the normalized form so no redundancy is present. The memory occupied by the query is 49%.

3.5 Star Vs Snowflakes Schema.

The example above for both of the schemas mention or clear the results according to various aspects. These results were accruing from performing an experiment on large data set and it can be said that these figures are accurately applicable dealing with large data sets. Below under the headings, the results compared to clear the reasons for the difference in results obtains.

Query Complexity.

It is evident that star and snowflake schema both perform differently due to their nature or properties. When it comes to query complexity, always snowflake schema is more complex for querying. Star schema has lesser number of tables because it is not in normalized form, which makes it easier to gather data from different tables, and in a snowflake, schema tables are normalized resulting in a greater number of tables and making it very much complex to query data. Complex joins are sometimes very difficult to handle and execute because a number of tables sometimes is far greater.

Execution time.

There is also a difference between their execution time. Star schema carries in it also the redundant data so traversing through that causes a bit time delay too, table size is also bigger because tables not split so traversal through the bigger table is a bit expensive. On the other hand snowflake schema is normalized tables are split carrying mostly relative data and no redundancy is there thus making the queries execute faster. Like according to the above example star schema takes 21s whereas snowflake schema takes 17s for execution. Therefore, for large data sets, star schema always takes more execution time than snowflake schema.

Effect on the size of data warehouse

Normalization is again a key role in discussing the size of data warehouse size. Star being the one having DE normalized dimensions will always occupy larger space because it will have a lot of redundant data hence requiring for space and memory and snowflake schema being completely normalized will require less space and memory than star schema. As mentioned in the table snowflake used 0.61 GB and star schema used 0.78 GB.

Results.

When we apply star schema, execute the query and get results we noted that the results of snowflakes have some difference from star schema. The query applies in Table: 6 and Table: 3. They show clearly the query execution time and resource consuming for the same data set.

Query Optimization

Star schema's performance figures gathered from the above-carried example are not much prompting to use them in the construction of our data warehouse. The performance of star schema can be improved by adding bitmap indexing. Performance of

snowflake can also be enhanced but comparing both snowflakes already have good figures but the example can be carried for both the schemas [17].

4 Conclusion

In dimension modeling, both star and snowflake schema are major roots that can be considered for the construction of a data warehouse. Everything that we have discussed in detail through the examples it is clear that both of the schemas have their own significant importance. At the time there are two categories like in simple execution snowflake schema is always going to have an upper hand due to the factors discussed above, but if some of the optimizations are, apply to star schema it can also consider efficient, like using or adding bitmap indexes to it. The question above cannot really be considered important because all of the use of snowflake and star schema lies in the fact that you need or your requirements. Because the examples that are discussed above are discussed over some concrete results which specify that whether you consider or give more importance to the size of a data warehouse or you want to maintain the other factors more efficient that is up to you. So in the start snowflake was looking a bit more to consider but it's not true even star schema can be made better with applying optimization techniques.

5 Future Work

In future, some other effective ways can be considered to make these schemas more optimized to be used hence giving a completely new dimension to the data warehouse field, which can be more beneficial for the companies to perform business intelligence.

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